A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Water treatment, conservation & new sources - Set 1

Mark scheme

AQA spec 3.2.2.4-3.2.2.5; The physical environment; textbook pp. 122-127

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Do not write outside the box

Question	Marking guidance	Marks
01.1	**Any 4 of 6** 1 mark each for any 4 of the 6 points below, in any order, maximum 4. **1 Topography:** a narrow exit from a large deep basin so a small dam holds a large volume **1 Geology:** impermeable rock beneath (no seepage), strong enough to support the dam, no faults / seismic activity **1 Catchment area:** large enough with regular, reliable rainfall / river inflow – climate not too hot (evaporation losses) **1 Catchment land use:** low pollution risk (industry, pesticides) – low soil erosion (sedimentation reduces capacity) **1 Infrastructure / proximity to demand:** access for construction, cost of transporting water **1 Existing land use:** value of what is lost (urban areas, conservation sites, farmland) - loss:benefit analysis **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **impermeable** → does not let water through • **reliable** → repeatable / gives consistent results when repeated **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself	4
01.2	**Marks shared** Up to 3 marks in total - share them between the 2 points below according to how completely each one is made. **up to 2 Advantage:** fewer land-use conflicts as no land is flooded – large volume near coastal cities **up to 2 Disadvantages:** destroys / changes important intertidal habitats (mudflats for wading birds) – obstacle to shipping – collects pollution from the whole river catchment – sedimentation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	3
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** Very energy-intensive and expensive (reverse osmosis / distillation) **1** Only worthwhile where seawater is available and freshwater supplies are inadequate (eg Malta, Middle East) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

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Question	Marking guidance	Marks
01.4	1 mark for the point below. **1** Surplus water in wet periods (when infiltration capacity is exceeded and runoff would go to the sea) is pumped underground or diverted into lagoons to infiltrate, so abstraction in dry periods is balanced by recharge **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquifer** → a permeable rock that stores groundwater / underground water store • **sustainable** → meeting present needs without stopping future generations meeting theirs • **infiltration** → water soaking into the soil surface **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-124; AQA spec 3.2.2.5.	
02.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** (150 - 125) / 150 x 100 = 16.7% (accept 17%) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** A fixed fee gives no incentive to save water because the bill is the same however much is used **1** Metering matches the charge to the volume used so households save money by using less (fewer hosepipes, low-flow appliances) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

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Question	Marking guidance	Marks
02.3	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1 Low-water appliances:** dual-flush toilets, low-flow shower heads, efficient washing machines / dishwashers, tap timers **1 Greywater re-use:** bath / shower water used to flush toilets or water gardens **1** Rainwater collection / water butts for garden use **1 Xeriscaping:** planting drought-adapted species that need no watering **1** Repairing leaks / dripping taps **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
02.4	**1 mark each** 1 mark for each of the 2 points below. **1** Old infrastructure / pipes in poor condition leak **1** Repair and good maintenance / pipe replacement / leak detection reduce wastage **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
02.5	**1 mark each** 1 mark for each of the 2 points below. **1** Water is delivered directly to the plant roots / soil around each plant **1** Less evaporation loss from spray droplets and from wetting the whole soil surface – less runoff **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 124-126; AQA spec 3.2.2.5.	
03.1	**1 mark each** 1 mark for each of the 3 points below. **1** Clay particles are very small and carry electrostatic charges on their surfaces so they repel each other and stay suspended **1** Flocculants (aluminium sulfate / alum or polyelectrolytes) are added and mixed to neutralise the charges **1** The particles clump together (flocs) and settle in the clarifier tank **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** Increases dissolved oxygen and removes hydrogen sulfide (bad-egg smell) from anaerobic sources **1** Oxidises some dissolved metals (eg iron) so they become insoluble and can be removed – improves taste **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** Peaty water contains organic matter which reacts with chlorine to produce toxic substances (eg trihalomethanes) **1** Ozone / UV kill pathogens without forming these by-products (though chlorine / chloramine keeps water sterile during distribution) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
03.4	**1 mark each** 1 mark for each of the 3 points below. **1 Ion exchange:** toxic ions adsorb onto polymer resin beads or zeolite (also used to soften water by removing calcium / magnesium) **1** Reverse osmosis / distillation (desalination) - expensive, energy-intensive **1 Activated carbon:** organic molecules causing odour / taste and pesticides adsorb onto the high-surface-area carbon particles **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **adsorb** → nutrient ions sticking to the surfaces of clay and humus **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 126-128; AQA spec 3.2.2.4.	
04.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - rivers are accessible over a large area with few natural contaminants	1

Do not write outside the box

Question	Marking guidance	Marks
04.2	**1 mark each** 1 mark for each of the 2 points below. **1 Potable / public supply:** no pathogens, low toxins, good appearance / taste / smell (public health) **1 Industrial boiler / cooling water:** soft water with low calcium so mineral deposits do not block pipes or reduce heat exchange – spray irrigation: low turbidity to avoid blocked pipes and low heavy metals **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	2
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** Canals / pipes move water from areas with a surplus to areas with a shortage (eg Wales to England, China's Yangtze transfers) **1 Disadvantage:** high cost / energy for pumping – ecological impacts on the donor river – transfer of pollutants or invasive species – political conflict **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-128; AQA spec 3.2.2.4-3.2.2.5.	
05.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 A:** Its total annual discharge	1
05.2	**1 mark each** 1 mark for each of the 2 points below. **1** Public supplies cannot match consumer demand / the water is often cleaner than rivers **1** It also reduces urban flooding after heavy rain – surplus can be used for aquifer recharge **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquifer** → a permeable rock that stores groundwater / underground water store	2

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Question	Marking guidance	Marks
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** The reservoir is created in the estuary so no farmland / settlements need to be flooded **1** But it changes important inter-tidal habitats, obstructs shipping and can be polluted by activities anywhere in the river's catchment **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-123; AQA spec 3.2.2.5.	
06.1	**1 mark each** 1 mark for each of the 4 points below. **1 Potable:** no pathogens, toxins at acceptable levels, looks / smells / tastes good - public health, consumer complaints **1 Spray irrigation:** low turbidity, low toxins such as heavy metals - blockage of pipes, contamination of food **1 Condenser cooling:** no gross solids - pipe blockage **1 Boiler water:** no dissolved minerals / soft water - mineral deposits reduce heat exchange and block pipes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'the body breaks the metal down' - metals are elements, cannot be broken down, and are stored in tissue	4
06.2	**1 mark each** 1 mark for each of the 2 points below. **1** Raises dissolved oxygen and removes hydrogen sulfide (bad-egg smell) from decay of organic matter **1** Some dissolved toxic / bad-tasting metals become insoluble when oxidised and can be removed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
06.3	**1 mark each** 1 mark for each of the 2 points below. **1** Clay particles carry electrostatic charges that make them repel each other so they stay suspended **1** Flocculants such as aluminium sulfate (alum) or polyelectrolytes neutralise the charges so the particles clump and settle in the clarifier **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

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Question	Marking guidance	Marks
06.4	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** Chlorine **1** Ozone **1** Ultraviolet light **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 126-127; AQA spec 3.2.2.4.	
07.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** No reservoir range = 3.9 - 0.7 = 3.2 m – with reservoir = 2.7 - 1.7 = 1.0 m **1** Reduction = 2.2 m **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (m) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working. • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2
07.2	**1 mark each** 1 mark for each of the 2 points below. **1** After heavy rainfall surplus water is stored, reducing peak flows / flooding downstream **1** During dry periods the dam is opened to release water and maintain river levels **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2

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Question	Marking guidance	Marks
07.3	**1 mark each** 1 mark for each of the 2 points below. **1** Deforestation / urbanisation reduce infiltration and interception **1** So runoff reaches the river faster after rain and less water is stored to maintain flow in dry periods **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **infiltration** → water soaking into the soil surface **Do not credit** • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	2

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Question	Marking guidance	Marks
07.4	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Site factors:** topography - narrow exit from a large deep basin so a small dam holds a large volume – impermeable, strong bedrock without faults or seismic activity – large catchment with regular high rainfall and low evaporation – catchment land use with low pollution risk (pesticides, industry) and low soil erosion to avoid sedimentation – forest / vegetation in the flooded area decaying anaerobically releasing methane – access for workers and materials and distance to demand – value of the land lost (urban, conservation, agricultural) - loss-benefit analysis differs between countries • **Alternatives for supply:** rainwater collection, rivers, estuary barrages, desalination (energy-intensive so only where freshwater is inadequate), unexploited aquifers (N Africa, Kenya), inter-basin transfers (China, Australia, Wales-England), artificial aquifer recharge using surplus wet-season water, river-regulation reservoirs, afforestation to even out flow • **Conservation strategies:** low-water appliances, dual-flush toilets, xeriscaping, drip irrigation, grey-water re-use, leak repair (about 20% lost in the UK), metering • Treatment / pollution control allowing re-use of contaminated sources: sewage treatment, buffer strips, mine drainage control • **Evaluation:** reservoirs give large reliable supplies and flood control but flood land, cause habitat loss and methane – conservation is cheap and quick but limited by behaviour – transfers create conflicts between regions – desalination has high energy / carbon cost – combination needed as demand grows • Justified conclusion	9

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Question	Marking guidance	Marks
	Also accept • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **sustainability** → meeting present needs without stopping future generations meeting theirs • **impermeable** → does not let water through • **aquifer** → a permeable rock that stores groundwater / underground water store **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself	
	Model answer - Level 3 (8 marks)	
	A reservoir site is judged first on topography: a narrow exit from a large, deep basin lets a small, cheap dam hold a large volume. The bedrock must be impermeable so water is not lost and strong enough, without faults or seismic activity, to carry the dam. The catchment should be large with regular, high rainfall and low evaporation, and its land uses should not pose a pollution risk from industry or pesticides or cause the soil erosion that would silt the reservoir. Flooding forest or vegetation causes anaerobic decay and methane. Access for workers and materials, distance to the area of demand, and the value of the land lost - urban areas and nature reserves protected, farmland less so - complete a loss-benefit analysis that different countries weigh differently.	
	Supply can also be increased without new reservoirs: rainwater collection in cities, estuary barrages, desalination where freshwater is genuinely inadequate, newly found aquifers in North Africa and Kenya, inter-basin transfers as in China and from Wales to England, artificial recharge of aquifers with wet-season surplus, river-regulation reservoirs and afforestation to even out flow. Demand can be cut by low-water appliances, xeriscaping, drip irrigation, grey-water re-use, metering and repairing the leaks that lose about a fifth of UK supply, and treatment lets contaminated sources be re-used.	
	Reservoirs give large, reliable supplies and flood control but flood land and habitats and release methane; conservation is cheap and quick but limited by behaviour; transfers create conflicts between regions; desalination carries a high energy and carbon cost. As demand grows, the sustainable answer is conservation first, then a combination of sources matched to each region.	
	Weaker answer - Level 2 (4 marks)	
	A reservoir needs a large catchment area with lots of rain, impermeable rock so the water does not drain away, and a narrow valley so the dam is small. It should be near the people who need the water and not flood important land. Other ways to get water include desalination, which is expensive, and collecting rainwater. Water can be conserved by fixing leaks, using water meters and having low-flush toilets. All of these help make water supplies more sustainable.	
	What makes the difference	

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Question	Marking guidance	Marks
	• Level 2 lists correct site factors but without the reasons (why impermeable, why the valley shape matters, why land use in the catchment matters, methane from flooded vegetation).	
	• Level 3 covers far more alternatives (aquifer recharge, river regulation, transfers, afforestation) and evaluates each; Level 2 evaluates only desalination.	
	• Level 3 ends with a prioritised judgement; Level 2 ends with 'all of these help'.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 122-127; AQA spec 3.2.2.4-3.2.2.5.	

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